Environmental Research 257 (2024) 119410

Contents lists available at ScienceDirect

Environmental Research

journal homepage: www.elsevier.com/locate/envres

Rainfall seasonality shapes microbial assembly and niche characteristics in Yunnan Plateau lakes, China

Qi Liu [a], Xinlu Duan [a], Yang Zhang [a], Lizeng Duan [a], Xiaonan Zhang [a], Fengwen Liu [a], Donglin Li [a], Hucai Zhang [a,b,*]

[a] Institute for Ecological Research and Pollution Control of Plateau Lakes, School of Ecology and Environmental Science, Yunnan University, Kunming, Yunnan, 650500, China

[b] Southwest United Graduate School, Kunming, 650500, Yunnan, China

ARTICLE INFO

Keywords:
Determinism
Environmental heterogeneity
Microbial diversity
Microbial assembly
Niche breadth
Plateau lake

ABSTRACT

Microorganisms are crucial components of freshwater ecosystems. Understanding the microbial community assembly processes and niche characteristics in freshwater ecosystems, which are poorly understood, is crucial for evaluating microbial ecological roles. The Yunnan Plateau lakes in China represent a freshwater ecosystem that is experiencing eutrophication due to anthropogenic activities. Here, variation in the assembly and niche characteristics of both prokaryotic and microeukaryotic communities was explored in Yunnan Plateau lakes across two seasons (dry season and rainy season) to determine the impacts of rainfall and environmental conditions on the microbial community and niche. The results showed that the environmental heterogeneity of the lakes decreased in the rainy season compared to the dry season. The microbial (bacterial and microeukaryotic) α-diversity significantly decreased during the rainy season. Deterministic processes were found to dominate microbial community assembly in both seasons. β-Diversity decomposition analysis revealed that microbial community compositional dissimilarities were dominated by species replacement processes. The co-occurrence networks indicated reduced species complexity for microbes and a destabilized network for prokaryotes prior to rainfall, while the opposite was found for microeukaryotes following rainfall. Microbial niche breadth decreased significantly in the rainy season. In addition, lower prokaryotic niche overlap, but greater microeukaryotic niche overlap, was observed after rainfall. Rainfall and environmental conditions significantly affected the microbial community assembly and niche characteristics. It can be concluded that rainfall and external pollutant input during the seasonal transition alter the lake environment, thereby regulating the microbial community and niche in these lakes. Our findings offer new insight into microbiota assembly and niche patterns in plateau lakes, further deepening the understanding of freshwater ecosystem functioning.

1. Introduction

Microbiomes play a crucial role in maintaining a variety of ecosystem functions and services within freshwater ecosystems, such as primary production, nutrient cycling, carbon cycling, and the decomposition of organic matter (Bardgett and van der Putten, 2014; Liu et al., 2023a; Montoya et al., 2006). Analysis of microbial assembly and niches across environmental gradients can help disentangle the processes that drive microbial communities and ecological dynamics (Laland et al., 2016; She et al., 2021). Environmental determinism, such as changes in environmental conditions, and ecological stochasticity, such as dispersal limitation, have been reported to regulate microbial community assembly in different ecosystems (Dini-Andreote et al., 2015; Stegen et al., 2012; Zhou and Ning, 2017). These processes shape niche partitioning, allowing ecologically similar organisms to coexist under conditions of limited resources (Kartzinel et al., 2015). Recent findings have provided exciting perspectives on microbial diversity, assembly patterns, and niche differences in soil microbes (Delgado-Baquerizo et al., 2016), root-associated microbiomes (Ofek-Lalzar et al., 2014), and biofilm-forming marine microorganisms (Zhang et al., 2019). Additionally, the impact of microplastic pollution on the microbial community in freshwater and seawater ecosystems and the importance of

* Corresponding author. Institute for Ecological Research and Pollution Control of Plateau Lakes, School of Ecology and Environmental Science, Yunnan University, Kunming, Yunnan, 650500, China.

E-mail address: zhanghc@ynu.edu.cn (H. Zhang).

https://doi.org/10.1016/j.envres.2024.119410
Received 26 March 2024; Received in revised form 5 June 2024; Accepted 10 June 2024
Available online 11 June 2024
0013-9351/

environmental factors for microbial community assembly and distance–decay relationships in water and sediment niches have been evaluated in aquatic environments (Gu et al., 2023; Li et al., 2021). However, current understanding of the assembly processes and niche characteristics of microbial communities in freshwater ecosystems is limited.

Freshwater ecosystems provide vital ecosystem services for both nature and society (Albert et al., 2021). Lakes are one of the most important components of freshwater ecosystems, holding much of Earth's accessible liquid freshwater and supporting biodiversity and ecosystem functions around the world (Woolway and Merchant, 2019). Lake environments are vulnerable to abrupt climate changes, such as altered rainfall regimes, and anthropogenic activities, such as water pollution and the expansion of agricultural and urban landscapes (Cavicchioli et al., 2019; Dudgeon, 2019). Changes in lake ecosystems can significantly disturb the microbial ecosystems of prokaryotes and microeukaryotes. For example, acid stress has been shown to drive the community assembly and species coexistence of microbiota in acid mine drainage lakes (She et al., 2021). Water transparency, nutrient concentration, and fish grazing were suggested to be the main driving forces in determining bacterioplankton community composition in freshwater lakes (Cao et al., 2017; Chen et al., 2023).

The Yunnan Plateau (southwestern China), located in a subtropical or temperate climate zone bordering the Tibetan Plateau to the northwest (Zhang et al., 2015), is one of five regions with the richest natural lakes in China (Ma et al., 2011; Ran et al., 2023) and is strongly affected by the South Asian summer monsoon (Zhang et al., 2021a). Therefore, this region has a distinct dry season and rainy season (Liu et al., 2023a). Under the distinct climatic conditions of the dry and rainy seasons, the microbial composition and function exhibited distinct characteristics. For example, our previous study on Yunnan Plateau lakes showed that environmental effects, notably trophic state and rainfall, are correlated with both microbial community structure and the metabolic potential of microbes in the dry and wet seasons (Liu et al., 2023a), which is consistent with previous metagenomic studies (Shen et al., 2019). Another study showed that the microbial community structure responded to changes in nutrient composition between the dry season and wet season in Poyang Lake, China (Ren et al., 2019). The metabolic activity and functional diversity of the mangrove soil microbial community were found to be higher in the rainy season than those in the dry season (Wang et al., 2024). Compared with lakes in floodplains (such as Lake Taihu and Lake Chaohu in China), the Yunnan Plateau lakes are distributed in different geographical regions, are relatively separated (Liu et al., 2015), and are more sensitive to climate change and anthropogenic activities (Daly et al., 2019; Liu et al., 2023a). The altitude and solar ultraviolet (UV) radiation have been reported to influence microbial communities (Karlsson et al., 2001; Sommaruga, 2001), suggesting that the high altitude and UV radiation of the region may impact the microbiota.

The Yunnan Plateau lakes play a vital role in regulating the climate of watersheds, offering freshwater resources for adjacent urban and rural areas, and providing a tourist attraction and irrigation water for agriculture. However, their water quality has deteriorated considerably during the last few decades because of climate change and increasing human interference (Zhang et al., 2007, 2021b). Agricultural runoff has been identified as a key driver of water quality degradation on the Yunnan Plateau (Zhang et al., 2007). In addition, urbanization generally leads to increased nutrient loads in surrounding lakes (Kakouei et al., 2021). Built-up lands, such as urban areas, industrial areas, and large rural settlements, have been found to be the leading source of pollution on the Yunnan Plateau. Positive correlations between the percentage of built-up land in watersheds and the nutrient contents (i.e., nitrogen and phosphorous) of lakes have been detected (Liu et al., 2012). These factors, together with the unique climatic characteristics of the Yunnan Plateau, including distinct dry and rainy seasons, facilitate further investigations of the impacts of climate change and anthropogenic activities on microbial assembly and niches in freshwater ecosystems. Herein, changes in the assembly and niche characteristics (niche breadth and niche overlap) of prokaryotic and microeukaryotic communities were examined in Yunnan Plateau lakes in China across two seasons (i.e., the dry season and the rainy season). We aimed to (i) assess the variations in the microbial community assembly and niche characteristics across the two seasons and (ii) disentangle the intrinsic ecological mechanisms underlying the differences in the assembly processes and niches of the microbial communities. The data generated herein provide a comprehensive framework for evaluating the ecological status of lakes.

2. Materials and methods

2.1. Experimental design and sampling

This study focused on nine plateau lakes located on the Yunnan Plateau in southwestern China, each with an area greater than 30 km^2 and varying trophic statuses (Liu et al., 2023a). The region has a distinct dry season, which is mainly from late October to early May of the next year, and a rainy season, which lasts from late May to early October of each year (Liu et al., 2023a). The sampling points for each lake were evenly set based on the size of the lake area to represent the overall situation of each lake (Fig. 1). Water samples were collected in sterile 3 L glass bottles in each lake from the surface water (top 0.5 m) between May 2020 for the dry season (before the rainy season, when the lake level is lowest) and October 2020 for the rainy season (immediately after the rainy season, when the annual lake level is the highest). The quantity of water samples collected in the dry season and the rainy season was the same. The number of water samples taken from each lake is as follows: n = 11 in Lake Dianchi, n = 9 in Lake Fuxian, n = 8 in Lake Xingyun, n = 8 in Lake Chenghai, n = 8 in Lake Erhai, n = 8 in Lake Lugu, n = 7 in Lake Qilu, n = 6 in Lake Yilong, and n = 9 in Lake Yangzong. Three replicates were performed to analyze the microbial communities at each sampling site.

2.2. Analysis of environmental parameters

Water temperature (WT), dissolved oxygen (DO), and pH were measured *in situ* using a multiparameter probe (Professional Plus, Yellow Springs Instruments, USA). The concentrations of total nitrogen (TN) and total phosphorus (TP) in lake water were measured following standard methods (State Environmental Protection Administration of China, 2002). Water samples for the determination of chlorophyll *a* (Chl-*a*) were filtered through a Whatman GF/F filter (biomass intercepted by a 0.7-μm pore size). The filters were immersed in 90% acetone and extracted in the dark at 4 °C for 24 h. The extractant was measured using a spectrophotometer (TU-1810, DPC, China) and the concentration of Chl-*a* was then calculated following standard methods (Arar, 1997). The trophic level index (TLI) values were used to evaluate the trophic conditions of the nine lakes (i.e., oligotrophic, TLI ≤ 30; mesotrophic, $30 <$ TLI ≤ 60; eutrophic, TLI > 60) (Wang et al., 2019). The TLI is a weighted sum based on the correlations between Chl-*a* and other substances. The TN, TP, and Chl-*a* were used to calculate the TLI, and the corresponding formulas were established as follows:

$$TLI(TN) = 10(5.453 + 1.694\ \ln(TN)), \tag{1}$$

$$TLI(TP) = 10(9.436 + 1.624\ \ln(TP)), \tag{2}$$

$$TLI(Chl - a) = 10(2.5 + 1.086\ \ln(Chl - a)), \tag{3}$$

where the unit of Chl-*a* is μg L^{-1}; and the units for TN and TP is mg L^{-1}. The TLI was calculated using the following equations:

$$TLI = \sum_{j=1}^{m} W_j TLI(j), \tag{4}$$

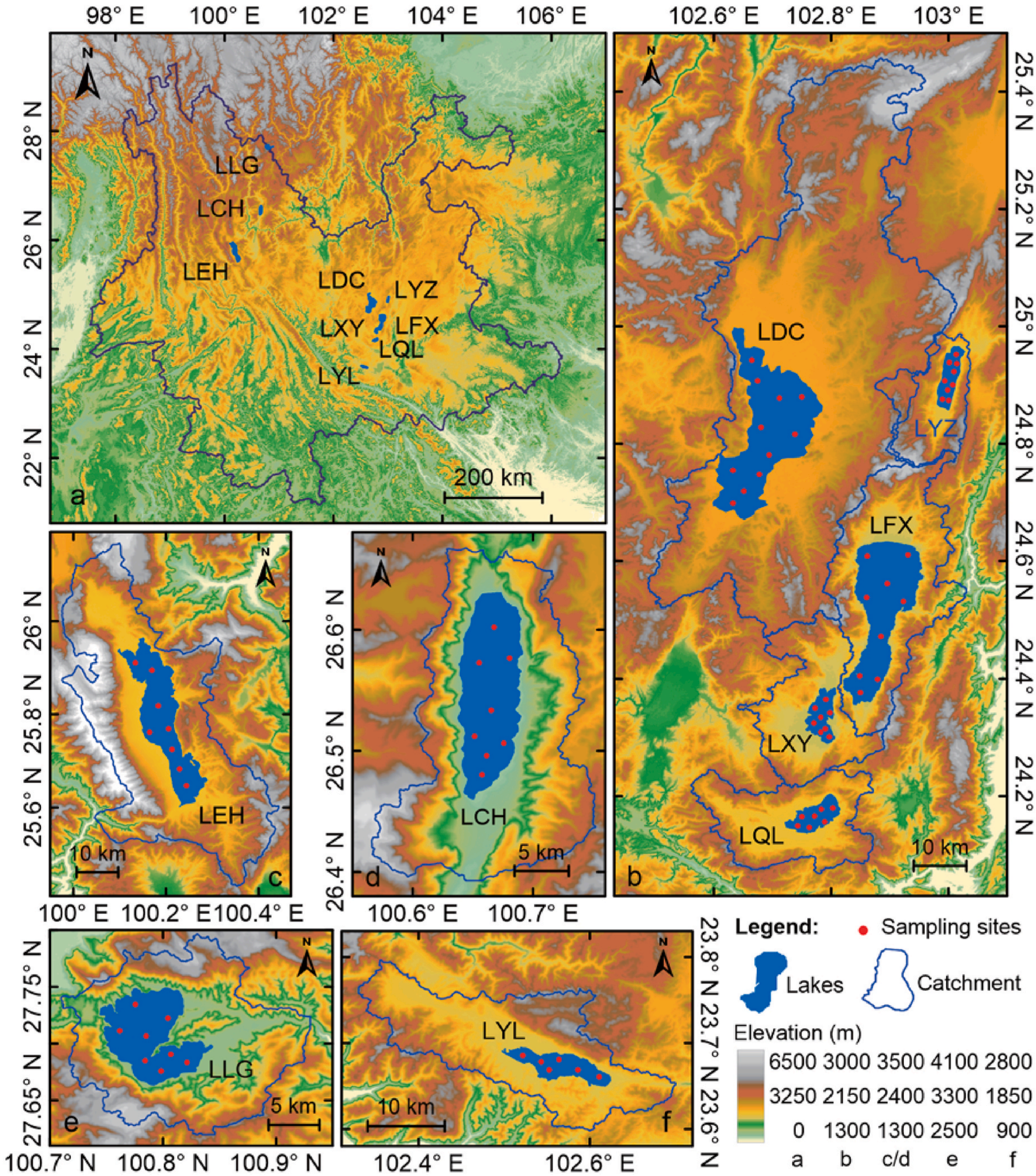

Fig. 1. Sample sites in lakes located on the Yunnan Plateau. LDC, Lake Dianchi; LYZ, Lake Yangzong; LFX, Lake Fuxian; LXY, Lake Xingyun; LQL, Lake Qilu; LEH, Lake Erhai; LCH, Lake Chenghai; LLG, Lake Lugu; LYL, Lake Yilong.

$$W_j = r_{ij}^2 \Bigg/ \sum_{j=1}^{m} r_{ij}^2, \tag{5}$$

where TLI(j) is the composite index of *j* with the correlative weight W_j; r_{ij} is the correlation coefficient between the reference Chl-*a* and each parameter *j* (Chl-*a*, 1; TP, 0.84; and TN, 0.82), and *m* is the number of indicators.

The rainfall data included the average rainfall at each sampling point in each lake, as described in our previous study (Liu et al., 2023a). Environmental heterogeneity (Ed) was evaluated by computing the average dissimilarity between sites (Ranjard et al., 2013) based on the six abiotic variables (i.e., WT, DO, pH, TN, TP, and TLI). For each season, Ed was calculated based on the Euclidean distance as follows:

$$Ed = \left(\frac{Euc}{Euc_{max}}\right) + 0.001, \tag{6}$$

where Euc is the Euclidean distance between two sites and Euc_{max} represents the maximum Euclidean distance considering all the pairwise distances in the overall dataset. A value of 0.001 was added to account for zero similarity between sites.

2.3. DNA extraction and metagenomic sequencing

The collected water samples were filtered through an MF-Millipore Membrane Filter (0.22 μm pore size, Millipore, Germany) using a vacuum filter, and the filtered membranes were placed at −80 °C until processing. A detailed description of DNA extraction and prokaryotic 16S rRNA and microeukaryotic 18S rRNA amplification can be found in our earlier publications (Liu et al., 2022a, 2023a). Briefly, DNA was extracted using the E.Z.N.A.® Water DNA Kit (Omega Biotek, Norcross, GA, U.S.) following the manufacturer's instructions. For prokaryotic community composition, 338F/806R primer sets (338F, ACTCCTACGGGAGGCAGCAG, 806R, GGACTACHVGGGTWTCTAAT) were used to amplify (in triplicate reactions for each sample) the 16S rRNA gene. For microeukaryotic community composition, the 3NDF/V4-euk-R2R primer pair (3NDF, GGCAAGTCTGGTGCCAG, V4-euk-R2R, ACGGTATCTRATCRTCTTCG) was selected to amplify (triplicate reactions for each sample) the 18S rRNA gene. The yields of the PCR products were quantified, purified, and then sequenced. Taxonomy was identified for each operational taxonomic unit (OTU) using the RDP classifier RDP Classifier algorithm (http://rdp.cme.msu.edu/) trained on the Silva (SSU138) 16S/18S rRNA databases for prokaryotic and microeukaryotic sequences. The prokaryotic and microeukaryotic compositions were identified at the phylum and genus levels (Fig. S1). For prokaryotes, Proteobacteria, Actinobacteriota, Cyanobacteria, and Bacteroidota were the dominant phyla in both seasons (Fig. S1a). The relative abundances of the main genera showed differences to various degrees, such as *CL500.29_marine_group*, *Acinetobacter*, and *Cyanobium_PCC.6307* (Fig. S1b). For microeukaryotes, SAR_k_norank, Cryptophyceae, and Arthropoda were the dominant taxa in both seasons and represented more than 50% of all phyla (Fig. S1c). At the genus level, *Cryptomonas*, *norank_o_Calanoida*, *norank_o_Ploimida*, and *norank_p_Phragmoplastophyta* were enriched in the nine plateau lakes (Fig. S1d). The raw sequencing data of the microbiota were published in our previous publication under accession numbers PRJNA783727 for Lake Lugu, PRJNA783733 for Lake Fuxian, PRJNA783742 for Lake Yangzong, PRJNA783751 for Lake Chenghai, PRJNA783756 for Lake Erhai, PRJNA783775 for Lake Dianchi, PRJNA783787 for Lake Xingyun, PRJNA783794 for Lake Qilu, and PRJNA783990 for Lake Yilong in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) (Liu et al., 2023a).

2.4. Data processing and statistical analyses

All analyses were performed in R version 4.2.1 (www.r-project.org). The nearest-taxon index (NTI) was calculated to investigate the phylogenetic community assembly at the within-community scale (Webb et al., 2002). A between-community null modeling approach was used to evaluate community assembly processes by calculating the β-nearest-taxon index (βNTI) using the "picante" package in R (Stegen et al., 2012; Zhou and Ning, 2017). Briefly, the phylogenetic distance between the closest relatives in two samples was calculated firstly using the beta mean nearest taxon distance (βMNTD). The βNTI is the fold of standard deviations between the observed βMNTD and the mean βMNTD of null distribution. The mean βNTI in combination with Bray–Curtis-based Raup-Crick (RC_{bray}) was used to distinguish the ecological processes, including deterministic processes (homogeneous selection and heterogeneous selection) and stochastic processes (homogeneous dispersal, dispersal limitation, and undominated processes), as follows: homogeneous selection ($\beta NTI < -2$), heterogeneous selection ($\beta NTI > 2$), homogenizing dispersal ($|\beta NTI| < 2$ and $RC_{bray} < -0.95$), dispersal limitation ($|\beta NTI| < 2$ and $RC_{bray} > 0.95$), and undominated processes ($|\beta NTI| < 2$ and $|RC_{bray}| < 0.95$) (Dini-Andreote et al., 2015; Stegen et al., 2012).

Microbial co-occurrence networks were constructed to estimate species coexistence in the nine lakes (She et al., 2021) using the "psych" package in R. Briefly, rare OTUs (relative abundance <1%) in each sample and OTUs that occurred in less than 50% of samples were eliminated before correlation analysis. Robust correlations with Spearman's correlation coefficients (ρ) > 0.6 and false discovery rate-corrected *P*-values <0.05 were used to construct networks. Networks were visualized using the Gephi platform (Bastian et al., 2009). Different metrics have been used to investigate the node-level topological features of the microbial community, including degree (the number of connections for a particular node), betweenness centrality (the potential influence of a particular node on the connections of other nodes), closeness centrality (the average distance of the node to any other node), and negative/positive cohesions (the ratio of negative to positive associations between OTUs) (Berry and Widder, 2014; Herren and McMahon, 2017; Jiao et al., 2020). The within-module connectivity (Zi) and among-module connectivity (Pi) were used to identify the roles of the nodes in the network as follows: peripheral nodes ($Zi < 2.5$, $Pi < 0.62$); connectors ($Zi < 2.5$, $Pi > 0.62$); module hubs ($Zi > 2.5$, $Pi < 0.62$); and network hubs ($Zi > 2.5$, $Pi > 0.62$) (Deng et al., 2012; Guimera and Nunes Amaral, 2005).

The microbial α-diversity (i.e., the Shannon index and Simpson index) and community dissimilarity in the two seasons were calculated based on Bray–Curtis distances using the "vegan" package in R (Liu et al., 2023b). Compositional dissimilarities (β-diversity) were partitioned into similarity, replacement, and richness difference components (Podani and Schmera, 2011; Shen et al., 2020) using the "adespatial" package in R. Levins' niche breadth index (Geng et al., 2022) and niche overlap (Xiong et al., 2022) were computed to infer the intra- and interspecific relationships using the "spaa" package in R. Partial least squares-path modeling (PLS-PM) was performed to investigate the linkages among rainfall, nutrients (loadings included TN, TP, and TLI), environmental factors (loadings included WT, pH, and DO), α-diversity (loadings included Shannon index and Simpson index), β-diversity (loadings included similarity, replacement and richness difference), microbial assembly (loadings included NTI and βNTI), and microbial niche (loadings included niche breadth and niche overlap) based on mode A (Zhang et al., 2023). The partial correlation coefficients were computed using the "plspm" package in R. Student's *t*-test was used to compare the differences. If normality and/or homoscedasticity were not verified, a nonparametric test (Wilcoxon test) was conducted. A *P*-value <0.05 was considered significant.

3. Results

3.1. Changes in environmental variables and environmental heterogeneity

Analysis of the environmental variables revealed that the TN concentration (1.38 ± 1.15 mg/L in the dry season, 1.91 ± 1.67 mg/L in the rainy season, Student's *t*-test $P < 0.05$), pH (8.93 ± 0.30 in the dry season, 9.28 ± 0.49 in the rainy season, Student's *t*-test $P < 0.0001$), and rainfall (33.01 ± 4.36 mm in the dry season, 176.29 ± 22.39 mm in the rainy season, Student's *t*-test $P < 0.0001$) were significantly greater in the rainy season than in the dry season (Fig. S2). The WT (19.96 ± 2.2 °C in the dry season, 17.94 ± 1.57 °C in the rainy season, Student's *t*-test $P < 0.0001$) and DO concentration (8.24 ± 1.57 mg/L in the dry season, 7.28 ± 1.54 mg/L in the rainy season, Student's *t*-test $P < 0.001$) significantly decreased in the rainy season, while the TP levels showed no significant differences between the two seasons (0.07 ± 0.07 mg/L in the dry season, 0.09 ± 0.10 mg/L in the rainy season, Student's *t*-test $P = 0.29$). Based on the TLI results, three lakes (Lake Chenghai, Lake Yangzong, and Lake Erhai) were consistently in a mesotrophic state, and four lakes (Lake Dianchi, Lake Xingyun, Lake Qilu, and Lake Yilong) were consistently in a eutrophic state in both seasons (Fig. S3). Lake Fuxian and Lake Lugu were oligotrophic in the dry season. After rainfall, however, Lake Fuxian was identified as a mesotrophic lake, and Lake Lugu was oligo-mesotrophic. The average environmental dissimilarity between sites was calculated to determine the Ed. As shown in Fig. S4, environmental heterogeneity was significantly greater in the dry season ($Ed = 0.26 \pm 0.08$) than in the rainy season ($Ed = 0.22 \pm 0.11$) (Student's *t*-test $P < 0.05$).

3.2. Microbial diversity and assembly processes

The Shannon index and Simpson index were used to investigate the within-sample diversity (i.e., α-diversity) of the microbial communities (Fig. 2a and b). The α-diversity indices showed that the within-sample diversity significantly decreased in the rainy season compared with that in the dry season (Wilcoxon test P = 1e-05). The community dissimilarity of both prokaryotes and microeukaryotes was significantly lower in the rainy season than in the dry season (Fig. S5). The community assembly indices (i.e., NTI and βNTI) were used to evaluate the microbial assembly processes in the nine plateau lakes (Fig. 2c and e). The NTI in prokaryotes was significantly higher in the dry season than that in the rainy season (Fig. 2c), while the opposite trend was found for microeukaryotes (Fig. 2e). The βNTI results showed that deterministic processes dominated both the prokaryotic and microeukaryotic community assemblies ($|\beta NTI| \geq 2$). Deterministic processes (with homogeneous selection as the major portion) accounted for more than 50% of the assembly of prokaryotes in the dry season, increasing to more than 70% in the rainy season (Fig. 2d). Similarly, homogeneous selection was predominant for the assembly processes of the microeukaryotic community, which was also found to be more prevalent in the dry season (more than 90%) than in the rainy season (more than 70%) (Fig. 2f).

3.3. Compositional dissimilarities of prokaryotic and microeukaryotic communities

According to the results of the taxonomic β-diversity decomposition, prokaryotic community compositional dissimilarities were dominated by similarity (46.40%) in the dry season and by species replacement (Repl) (50.22%) in the rainy season. For microeukaryotes, Repl had the greatest contribution to compositional dissimilarities (63.07% and 57.92% in the dry season and the rainy season, respectively) (Fig. 3). Similarity represented the species shared between samples. Repl exhibited comparable effects on compositional dissimilarities to RichDiff, which represents richness difference processes. The ratio of Repl to the compositional dissimilarity (Repl/D) of the prokaryotic community was significantly higher in the rainy season compared with that in the dry season (mean values of Repl/D: dry season, 0.4799; rainy season, 0.6717), while the opposite was true for the microeukaryotic community (mean values of Repl/D: dry season, 0.8226; rainy season, 0.7662). The results obtained by disassembling taxonomic β-diversity indicated an opposite change in the trend of species replacement of prokaryotes and microeukaryotes.

3.4. Co-occurrence networks of the microbial communities

Microbial co-occurrence networks were constructed to visualize and characterize the differences in species coexistence (Fig. 4a and 5a). Compared with the dry season, the level of closeness centrality was lower in the rainy season for both prokaryotes and microeukaryotes, which may be associated with the modularity of the co-occurrence network. There were fewer prokaryotic negative/positive cohesions in the rainy season than in the dry season, while microeukaryotic negative/positive cohesions were significantly higher in the rainy season (Fig. 4b and 5b). Further details are shown in Table S1. Furthermore, the topological roles of nodes were defined by two parameters, namely within-module connectivity (Zi) and among-module connectivity (Pi) (Fig. 4c and 5c). The majority of taxa were identified as peripherals with most of their links inside their own modules. Module hubs, connectors, and network hubs are believed to indicate keystone species in microbial communities (Sun and Jing, 2023). Greater proportions of both module hubs and connectors were obtained in the rainy season compared with those in the dry season for prokaryotes. A total of 68 and 123 keystone species were obtained in the prokaryotic ecological networks in the dry and rainy seasons, respectively. *CL500-29_marine_group* (OTU 11578 in the dry season; OTU 1987, OTU 2505, and OTU 16446 in the rainy season) and *Acinetobacter* (OTU 236 and OTU 12182 in the dry season; OTU 2, OTU 7693, and OTU 7777 in the rainy season) were identified as connectors in the two seasons. The proportions of connectors were higher, while the proportions of module hubs were lower in the rainy season compared with those in the dry season for microeukaryotes. In the microeukaryotic networks, 43 and 47 keystone species were observed in the dry and rainy seasons, respectively. *Vampyrellidae* (OTU 9346 and OTU 7605 in the dry season; OTU 9346 in the rainy season) and *Chlorophyceae* (OTU 3736 in the dry season; OTU 8834 in the rainy season) were identified as module hubs. *Chlorophyceae* (OTU 342, OTU 2773, and OTU 3411 in the dry season; OTU 302, OTU 1950, OTU 2773, OTU 7914, OTU 8207, OTU 9148, and OTU 9171 in the rainy season) was identified as a connector in both seasons. No network hubs were observed in any of the samples collected from these lakes.

3.5. Variations in the niche metrics of the prokaryotic and microeukaryotic communities

Niche breadth and niche overlap are key metrics for describing intra- and interspecific relationships (Baraloto et al., 2012; Granot and Belmaker, 2020). Here, the values of niche breadth and niche overlap were investigated to assess ecological trends in the microbial communities (Fig. 6). For both prokaryotes and microeukaryotes, niche breadth was greater in the dry season than in the rainy season. The changes in niche breadth are related to species diversity (such as the community composition and presence–absence), and increasing diversity and niche width promotes resource use (Gilbert, 2012). These results are a complement to changes in microbial diversity and resource utilization. The niche overlap of prokaryotes was lower in the rainy season. In contrast, microeukaryotes possessed a higher niche overlap in the rainy season than in the dry season.

3.6. The influence path among external factors, microbial diversities, niche indices, and assembly

The results of the PLS-PM analysis revealed different influences of external factors on microbial communities and niche indices (Fig. 7). In the dry season, rainfall directly influenced the prokaryotic niche (positive, $r = 0.768$) and assembly (negative, $r = -0.019$) but also had indirect negative ($r = -0.483$) and positive ($r = 0.613$) effects on the α-diversity and β-diversity in the prokaryotic niche and assembly, respectively. Nutrient levels were found to have positive effects on α-diversity ($r = 0.515$) and β-diversity ($r = 0.381$) and indirectly influenced the prokaryotic niche and assembly via diversity. The environmental factors had direct positive influences on the prokaryotic niche ($r = 0.276$) and assembly ($r = 0.235$). In the rainy season, direct influences of rainfall, α-diversity, and β-diversity on the prokaryotic assembly were not found. Rainfall indirectly affected the prokaryotic assembly via nutrient levels and environmental factors. Prokaryotic diversities influenced the assembly by impacting the prokaryotic niche. For microeukaryotes, the results demonstrated that environmental factors had positive and negative influences on the microeukaryotic niche ($r = 0.264$) and assembly ($r = -0.125$) in the dry season, respectively. Nutrient levels had positive influences on microeukaryotic diversities ($r = 0.188$ for α-diversity, $r = 0.364$ for β-diversity) and had negative effects on microeukaryotic niche ($r = -0.180$) and assembly ($r = -0.093$). In addition to the direct impact of rainfall on the microeukaryotic assembly ($r = -0.570$), rainfall also indirectly influenced the assembly by subsequently affecting nutrient levels, environmental factors, α-diversity, and β-diversity. In the rainy season, the direct effect of rainfall on the microeukaryotic assembly was not observed. Rainfall influenced the microeukaryotic assembly via nutrient levels, environmental factors, diversity, and niche. Based on the results of the PLS-PM analysis, α-diversity and β-diversity directly affected the microbial niche in both prokaryotic and microeukaryotic communities. The direct effect of microbial niche on microbial assembly was obtained in both seasons.

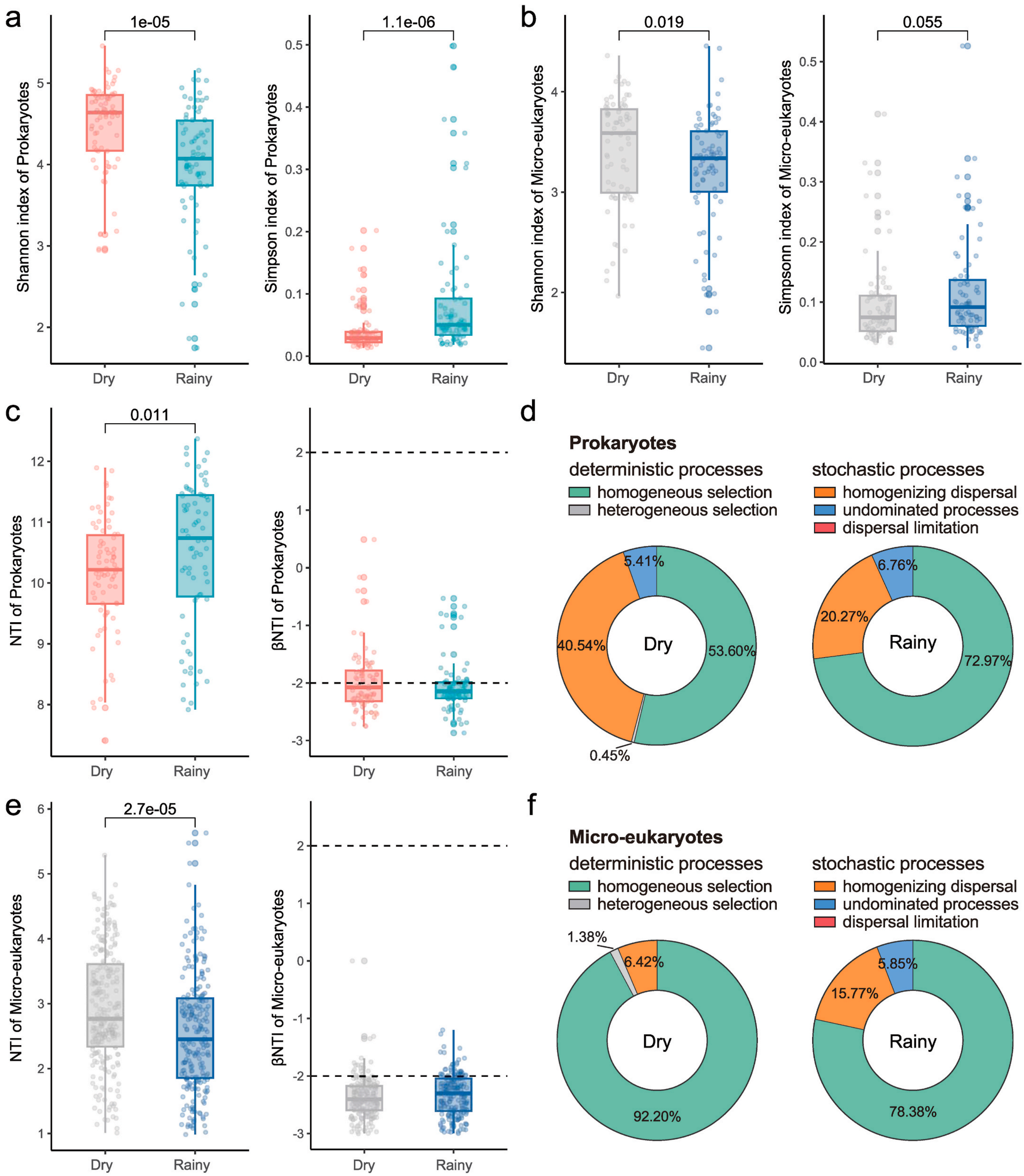

Fig. 2. The indices of α-diversity and assembly process analysis of prokaryotic and microeukaryotic communities in the plateau lakes between the dry season and rainy season. α-Diversity indices for prokaryotic (a) and microeukaryotic (b) communities. The NTI and βNTI values (c) and pie charts illustrating the assembly processes (d) of prokaryotic communities. The NTI and βNTI values (e) and pie charts illustrating the assembly processes (f) of microeukaryotic communities. The *P*-value was determined by a nonparametric test (Wilcoxon test).

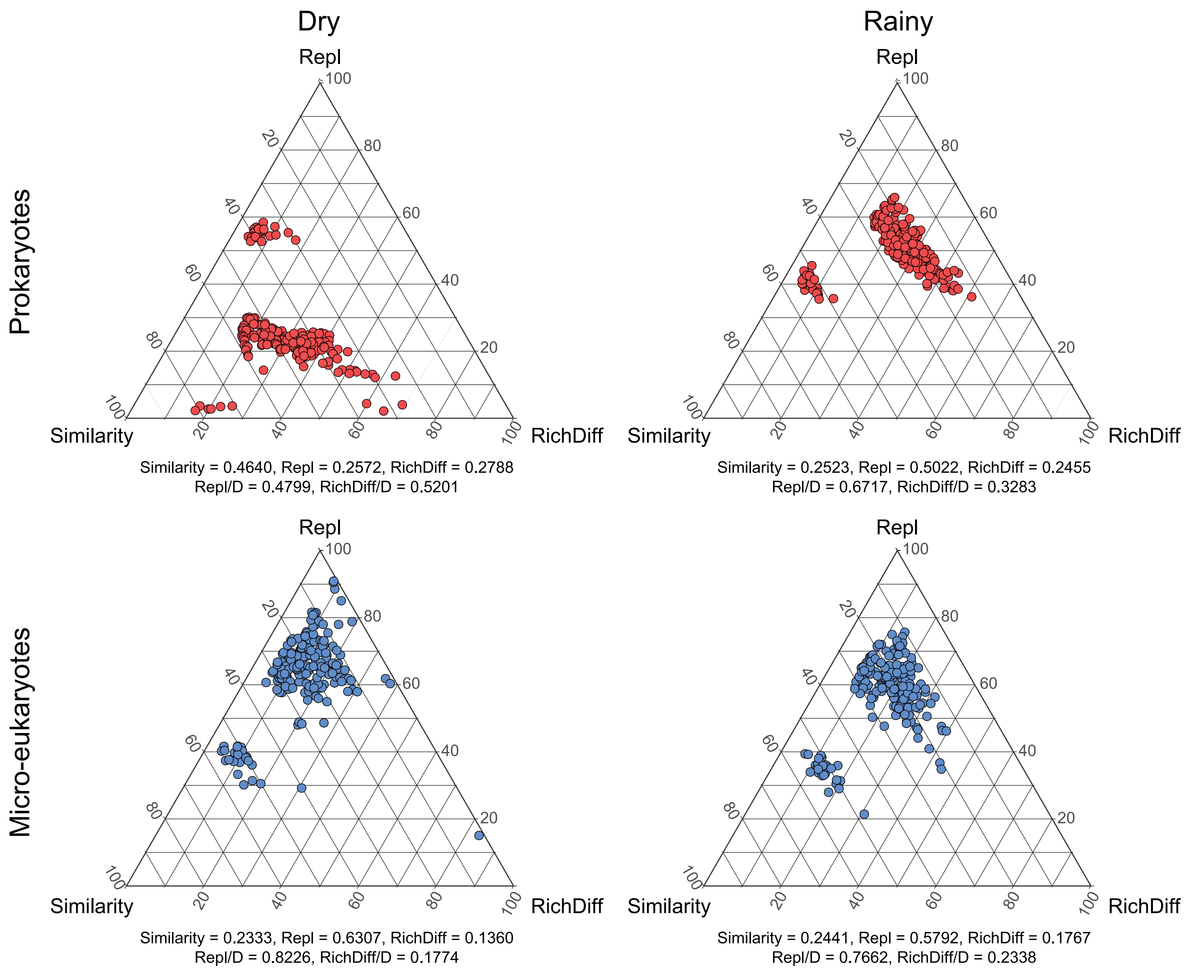

Fig. 3. Triangular plots of compositional dissimilarities for prokaryotic and microeukaryotic communities in the dry and rainy seasons. The position of each point is determined by a triplet of values from the Similarity = 1 − D (dissimilarity), Repl (species replacement), and RichDiff (richness difference) matrices. Each triplet is summed to 1. The mean values of Similarity, Repl, and RichDiff are shown.

4. Discussion

4.1. Rainfall potentially reduced environmental heterogeneity

Lake water quality has been shown to be influenced by various factors (e.g., natural factors and anthropogenic activities) at both the lake and watershed scales (Liu et al., 2012; Noges, 2009; Parker et al., 2008). The lower environmental heterogeneity in the rainy season than in the dry season in this study was likely due to increased agricultural runoff and nutrient loads from surrounding urban areas with organic matter and high loadings of TN and TP during the rainy season (Fig. S4) (Liu et al., 2012; Ren et al., 2019), which led to the lake water body tending toward a eutrophic state. The increased trophic status of lakes has been reported to decrease environmental heterogeneity through aggravating phytoplankton blooms over time (Wan et al., 2023). The nine Yunnan Plateau lakes studied here are all adjacent to cities. For example, Lake Dianchi is one of the most eutrophic lakes in China, and Kunming city is the largest city in the Yunnan Plateau and is located in the Lake Dianchi watershed (Wu et al., 2021). Lake Erhai is a mesotrophic plateau lake that is located in Dali city in Yunnan Province (Liu et al., 2022a), and Lake Xingyun is located in the Jiangchuan Basin in central Yunnan Province (Liu et al., 2022b). A previous study reported that the percentage of cropland in the watersheds of these nine lakes was approximately 25% (Liu et al., 2012). Extensive runoff from agricultural activities might have led to higher nutrient levels in the rainy season than in the dry season in these lakes (Chen et al., 2002; Ngoye and Machiwa, 2004). In addition, the increased nutrients in the water, decreased Ed, and high radiation levels on the Yunnan Plateau provide ideal conditions for cyanobacterial blooms in eutrophic lakes (Kakouei et al., 2021). As the Yunnan Plateau lakes are closed or semi-enclosed fault lakes (Zhang et al., 2018), they experience shortages of natural water resources and long water exchange cycles (Fan et al., 2023). This makes plateau lakes more sensitive to water quality deterioration, making it difficult to improve water quality. Therefore, we believe that the specific measures to protect and improve the water quality of plateau lakes should focus on optimizing the land-use type, developing intensive agricultural irrigation modes (Liu et al., 2012), controlling external nutrient loading and internal source pollution, and restoring the ecosystem. In addition, pollution during the rainy season is relatively higher not only because of rainfall but also because of differences in seasons and human activities, which may increase the homogeneity of the lake environments. Except for runoff during precipitation events, inflow rivers and anthropogenic inputs, such as wastewater treatment plant discharges and industrial discharges, may cause the accumulation and enrichment of nutrients and pollutants in lake water bodies and sediments (Wong et al., 2018; Xu et al., 2019). Substances (e.g., P) in lake sediments can affect lake ecosystems through endogenous releases with seasonal changes or differences in the hydrogeochemical parameters (Cai et al., 2023). More attention is needed to systematically analyze the effect of endogenous release from the sediments on freshwater lake systems. Moreover, reducing external loading is considered to benefit internal nutrient loading (Schindler, 2006; Wong et al., 2018).

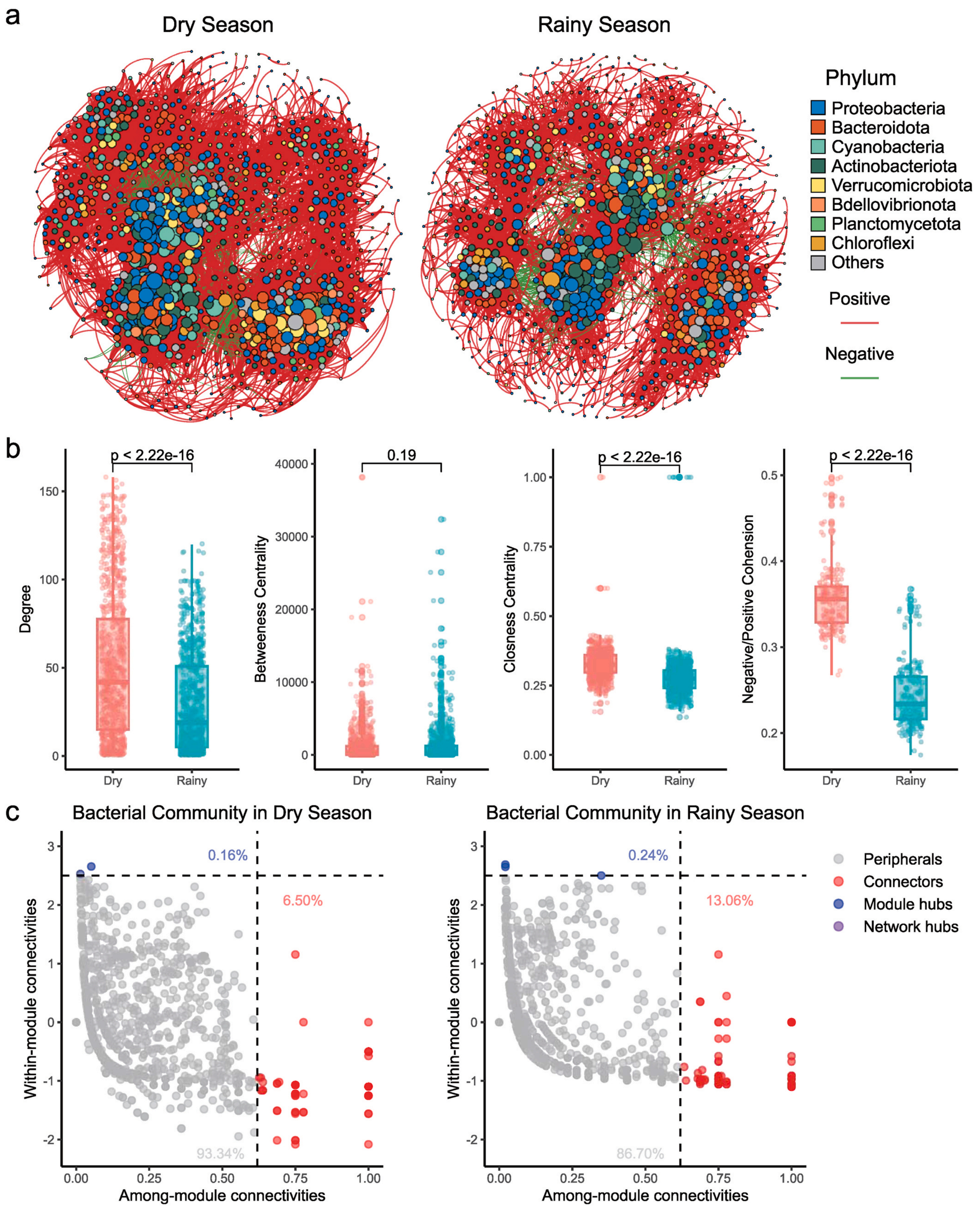

Fig. 4. Patterns of prokaryotic species coexistence. (a) Co-occurrence networks of prokaryotes in plateau lakes. The relative abundance is proportional to the size of each node, and the node color indicates the phylum level. The thickness of a connection between two nodes is proportional to the value of the Spearman's correlation coefficient. Red lines indicate positive correlations, while green lines indicate negative correlations. (b) Specific node-level network topological characteristics, including the degree, betweenness centrality, closeness centrality, and negative/positive cohesion, are shown. The *P*-values were determined using Student's *t*-test. (c) Topological roles of the network nodes, as determined by within-module connectivity (Zi) and among-module connectivity (Pi).

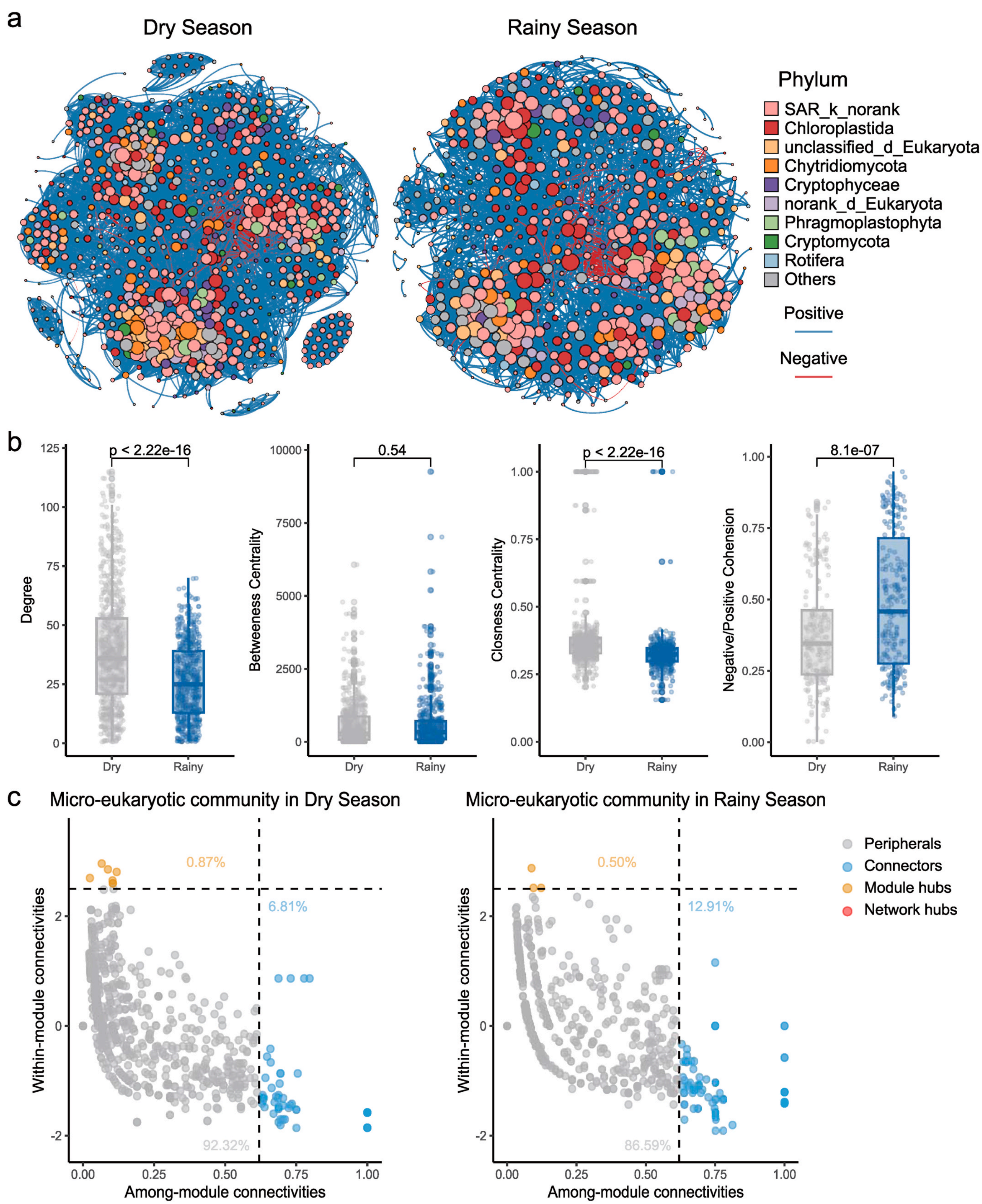

Fig. 5. Patterns of microeukaryotic species coexistence. (a) Co-occurrence networks of microeukaryotes in plateau lakes. The relative abundance is proportional to the size of each node, and the node color indicates the phylum level. The thickness of a connection between two nodes is proportional to the value of the Spearman's correlation coefficient. Blue lines indicate positive correlations, while red lines indicate negative correlations. (b) Specific node-level network topological characteristics, including the degree, betweenness centrality, closeness centrality, and negative/positive cohesion, are shown. The *P*-values were determined using Student's *t*-test. (c) Topological roles of the network nodes, as determined by within-module connectivity (Zi) and among-module connectivity (Pi).

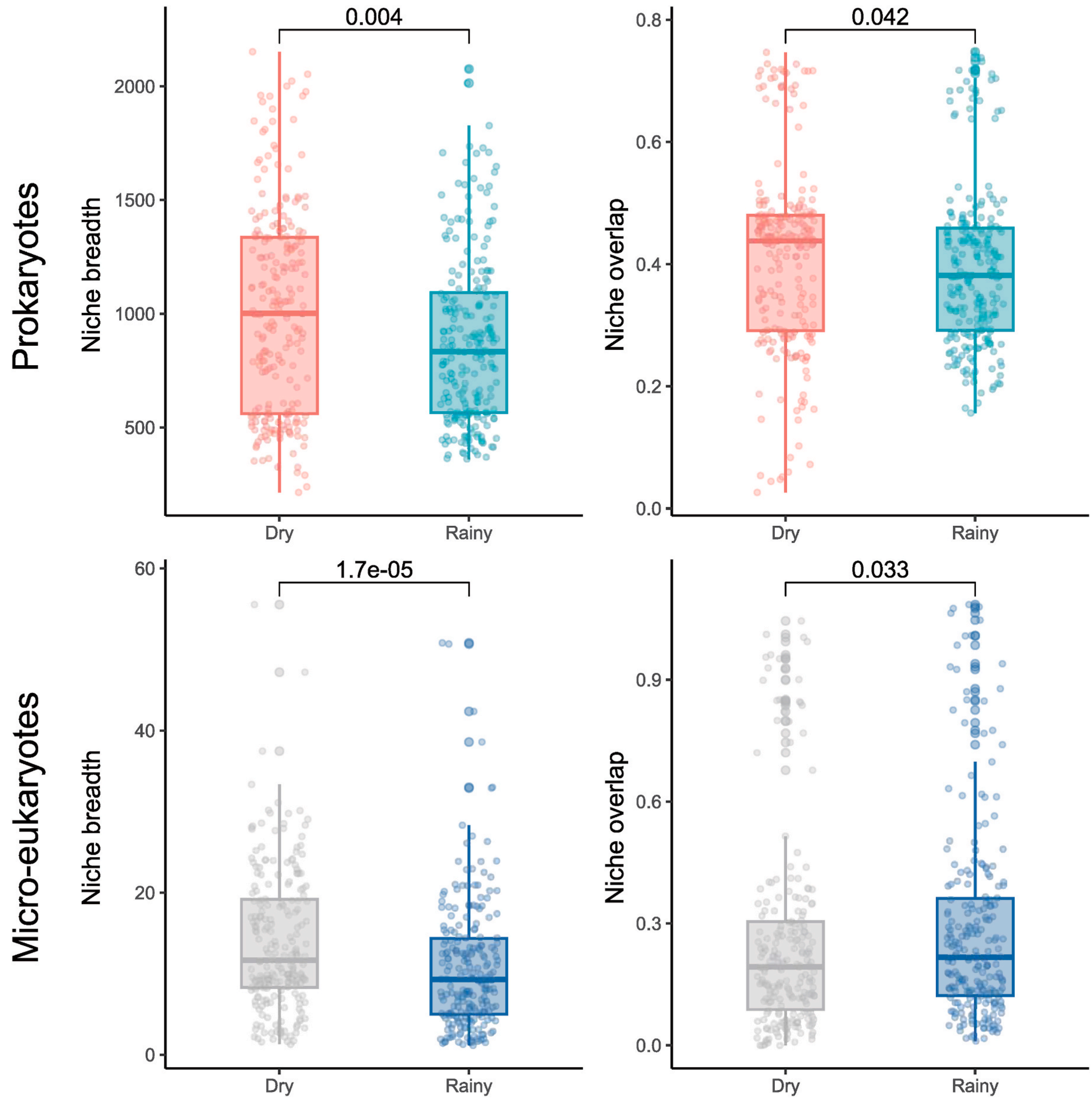

Fig. 6. Niche breadth and niche overlap analysis of prokaryotic and microeukaryotic communities in plateau lakes between the dry season and rainy season. The *P*-value was determined by a nonparametric test (Wilcoxon test).

Controlling anthropogenic eutrophication is also crucial for managing and conserving lake ecosystems.

4.2. Differences in the assembly processes of prokaryotes and microeukaryotes under environmental constraints

While changes in rainfall have been found to have a strong impact on microbial community structure and ecosystem functions (Su et al., 2022), most studies have focused on soil microbes (Manzoni et al., 2012; Ren et al., 2018; Zhang et al., 2016). We previously explored the differences in prokaryotic and microeukaryotic community compositions in each of the plateau lakes between the dry season and the rainy season (Liu et al., 2023a). For example, Proteobacteria, Actinobacteriota, Bacteroidota, and Cyanobacteria were identified as the major prokaryote phyla. Microeukaryotes were dominated by the phylum SAR_k_norank. Nutrient level-related factors (e.g., TN, TP, and TLI) and rainfall were found to be regulators of microbial communities. Other studies of lake microbes have also mainly focused on microbial community composition and diversity (Haukka et al., 2006; Liu et al., 2022a; Shen et al., 2019). However, the mechanisms of microbial assembly and ecological processes were not explored in these studies. Species replacement and abundance differences have been reported to explain the presence–absence and abundance of species in a community (Legendre, 2014). The β-diversity decomposition analysis revealed that microbial community compositional dissimilarities were dominated by species replacement (Repl) processes, and their respective contributions to this process differ. These results are consistent with earlier findings in lake sediments (Wang et al., 2023). However, the contributions of Repl to the dynamics of prokaryotic and microeukaryotic communities were different in lake water and sediments. Microeukaryotic communities were more affected by Repl in lake sediments (Wang et al., 2023). Our data indicated that Repl had a stronger effect on prokaryotic communities in the aquatic environment during the transition between the dry season and the rainy season (Fig. 3). The microbial turnover process is associated with the changes in their habitats, and a high level of microbial species replacement implies that environmental threats have stronger constraints on communities (Liang et al., 2015). A greater proportion of Repl in prokaryotic communities indicated a stronger

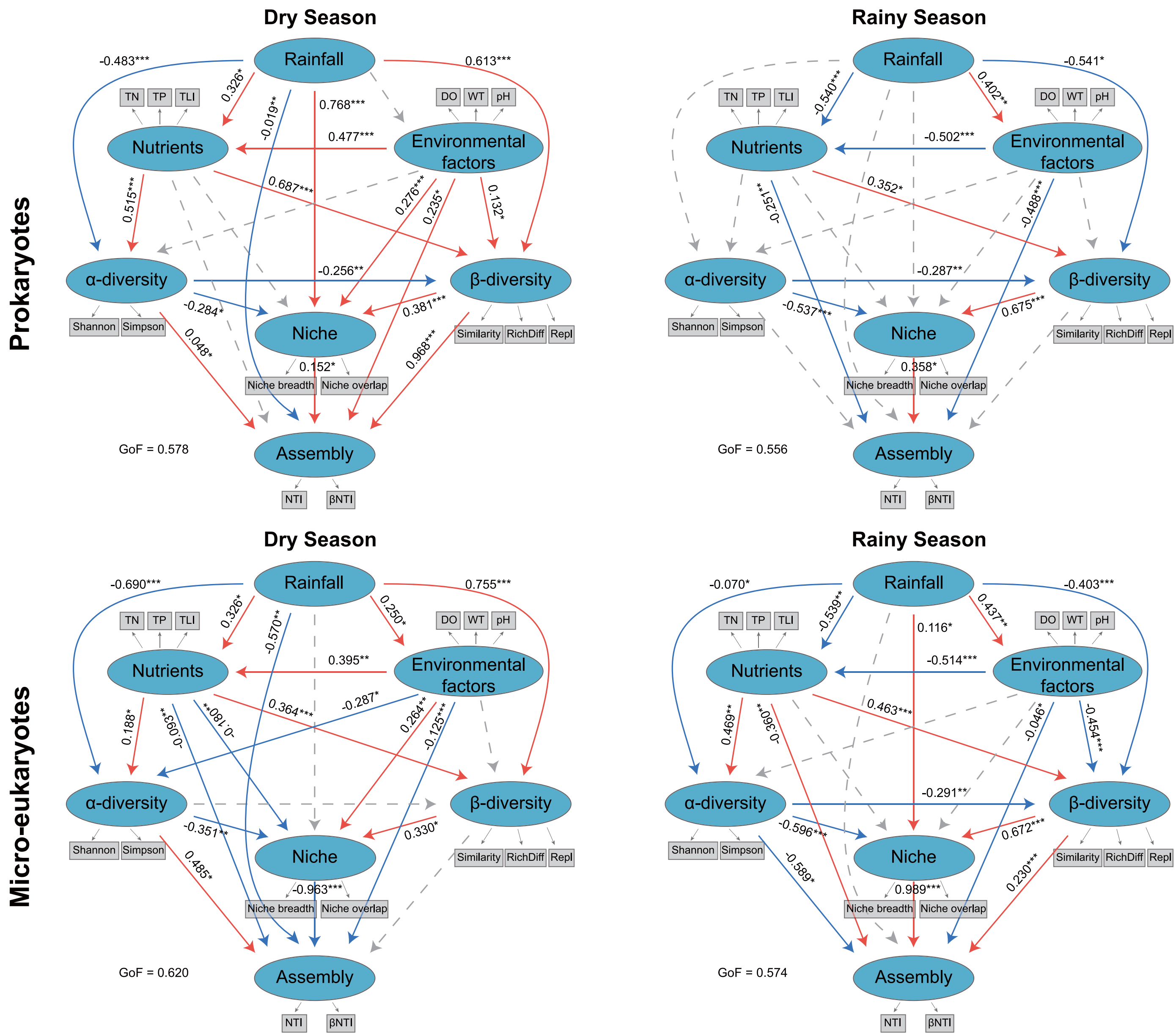

Fig. 7. Partial least squares-path models (PLS-PM) demonstrating the relationships among rainfall, environmental factors, nutrients, α-diversity, β-diversity, niche, and assembly of the prokaryotic and microeukaryotic communities.

environmental constraint on the prokaryotic community in freshwater environments. This suggests that investigations of lake sediments conducted for the purpose of lake management may not reflect changes in lake water because biodynamic processes vary in different habitats. In addition, rainfall and nutrients were found to directly affect the microbial β-diversity in both seasons. These results are the first to indicate that rainfall and nutrients may be the triggers leading to changes in microbial communities in plateau lake ecosystems, and these changes may be associated with reduced Ed caused by rainfall, along with the external input of pollutants during the seasonal transition.

Environmental heterogeneity not only reflects variations in environmental properties but is also considered to be an important factor influencing whether stochastic and/or deterministic processes dominate microbial community assembly (Huber et al., 2020). Both stochastic and deterministic processes balance microbial community assembly (Huo et al., 2022; Stegen et al., 2012). Here, deterministic processes dominated microbial community assembly in both seasons. However, the community assembly of prokaryotes and microeukaryotes showed opposite changes between the two seasons. Deterministic (54.05%) and stochastic (45.95%) processes balanced prokaryotic community assembly in the dry season, whereas deterministic processes (72.97%) had strong effects on the prokaryotic community assembly in the rainy season. For microeukaryotes, deterministic processes (93.58% for the dry season and 78.38% for the rainy season) dominated the community assembly during both seasons. Stochastic processes displayed limited effects in the dry season (6.42%) compared to the rainy season (21.62%) (Fig. 2). Based on the PLS-PM results, changes in environmental conditions had different degrees of influence on microbial community assembly (Fig. 7). The changes in Ed may result in higher deterministic processes in the rainy season for prokaryotes, which is consistent with a previous notion of bacterial metacommunity assembly in a floodplain river system (Huber et al., 2020). Different environmental conditions are thought to influence microbial communities by providing greater levels of resources and regulating assembly processes (Huo et al., 2022). When environmental conditions are homogeneous, homogeneous selection is the dominant ecological rule structuring prokaryotic communities

(Huber et al., 2020). Varying differentiation strategies between prokaryotes and microeukaryotes in response to changes in environmental heterogeneity were found in urban lakes in Wuhan, China as well (Wan et al., 2023). Decreased environmental heterogeneity was found to promote stochastic processes that determined microeukaryotic community assembly and deterministic processes that drove prokaryotic community assembly (Wan et al., 2023). In lotic habits, stochastic processes have been reported to be more important in shaping microbial community assembly (Yang et al., 2023). Flow regime affects the dispersal capability of the microbial community and makes stochastic processes predominate in these environments (Carrara et al., 2012). Deterministic processes were found to be the major mechanism structuring microbial communities in isolated lakes with low levels of hydrological connectivity, especially in plateau lakes. In addition, microbial lifestyle and cell size may also be related to different ecological processes dominating microbial community assembly (Spanbauer et al., 2016; Zinger et al., 2019). Prokaryotes are unicellular, while many microeukaryotes are typically regarded as multicellular. Several microeukaryotes generally have larger cell sizes than prokaryotes (Wan et al., 2023), and large-bodied organisms have been reported to be more stochastic than small-bodied organisms (Zinger et al., 2019). Moreover, the divergent responses of prokaryotes and microeukaryotes to environmental constraints may arise from the differences in dispersal rates between prokaryotes and microeukaryotes (Wan et al., 2023). Some microeukaryotes are able to obtain abundant nutrients with relative ease and survive in unfavorable conditions by actively moving in the water using feet, tails, and flagella (Wan et al., 2023). Although the strength of homogeneous selection was relatively high in comparison with homogenizing dispersal during the rainy season for microeukaryotes, stochasticity was observed to increase because of the high dispersal abilities and the insufficient environmental filtering as a selection force for applying species sorting under environmental homogeneity (Huber et al., 2020; Östman et al., 2012), which may have contributed to the increase in stochastic processes in the rainy season.

4.3. Response patterns of prokaryotic and microeukaryotic niches

Niche partitioning affects species coexistence in resource-limited environments (Kartzinel et al., 2015). The trophic network architectures between species, including niche breadth and niche overlap, play an important role in species interactions and the stability of the microbial community. Moreover, the characteristics of microbial niches have potential connections with deterministic processes and stochastic processes, which indicates that the niche is closely related to community assembly and may be linked to the relative contributions of fundamental ecological processes (Aslani et al., 2022; Fan et al., 2024; Gilbert, 2012). The niche breadth of a species describes the environment or resources it can inhabit or utilize (Gaston et al., 1997). Niche overlap represents the proportion of common resources used by two species (Ji and Wilson, 2002). Although microbial niche characteristics have been investigated in some urban lakes in floodplains (Wan et al., 2023), this aspect remains largely unknown in plateau lake ecosystems. In the lakes studied in this work, the niche breadth of both prokaryotes and microeukaryotes significantly decreased after rainfall (Fig. 6), indicating that more nutrients were available for microbial use during the rainy season. Nutrient inputs are considered to affect niche selection by providing more resources (Zhou et al., 2014). In heterogeneous environments, the species' utilization of resources is more complete due to extended resource use in species' niche spaces (Gilbert, 2012). The niche overlap results showed that prokaryotic niche overlap was lower in the rainy season than in the dry season, whereas microeukaryotic data increased in the rainy season. Dispersal potentials are believed to be associated with niche overlap among species (Gilbert, 2012). Positive correlations between microeukaryotic taxa, representing high niche overlap, may mediate the greater niche overlap during the rainy season (Hernandez et al., 2021). Our work provides novel insights into the changes in niche overlap between prokaryotes and microeukaryotes, indicating that the competitive intensity of prokaryotes for common resources was lower than that of microeukaryotes with the decrease of environmental heterogeneity (Hernandez et al., 2021; Huber et al., 2020). The variations in microbial niche indices may be related to changes in environmental conditions following rainfall, such as decreased Ed and increased nutrient levels, as confirmed by PLS-PM analysis. In addition, extensive runoff from the watersheds of these lakes may have provided additional resources for microbes. Furthermore, microbial diversity and lifestyle are also important microbial niche indices. The niche indices presented here are microbial characteristics (OTUs) that are related to microbial diversity (such as community composition and presence–absence) (Gilbert, 2012). Differences in species community and diversity can affect microbial resource utilization through associated metabolic potential (Liu et al., 2023a), thereby regulating the microbial niche through metabolic plasticity (Fan et al., 2024). Our previous study showed that metabolic pathways are more varied in prokaryotes following rainfall than in microeukaryotes (Liu et al., 2023a), allowing prokaryotes to utilize more diverse resources. As the main metabolic pathways of microeukaryotes are similar in the dry season and the rainy season, they tend to utilize common resources. The changes in microbial niches may be associated with different species interactions. In general, more positive relationships in the co-occurrence network indicate higher niche overlap, while negative relationships represent divergent niches (Hernandez et al., 2021). Negative/positive cohesion has been found to be a predictor of microbial network stability (Hernandez et al., 2021; Herren and McMahon, 2018; She et al., 2021). The causal relationships between microbial niches and co-occurrence relationships or network stability merit further study. The cell composition of microeukaryotes is generally more complex in structure and function than that of prokaryotes (Massana and Logares, 2013), improving their adaptation to ecological niches and weakening environmental filtering effects (Chen et al., 2022). Moreover, the active movement of microeukaryotes across habitats allows them to converge in a favorable environment, find resources, and evade predators, thereby playing a crucial role in allowing them to occupy relatively high trophic levels in aquatic environments (Wan et al., 2023; Wisnoski and Lennon, 2023), which may be responsible for the increased niche overlap under environmental constraints.

In lake ecosystems, ecosystem services provided by microbes (prokaryotes and microeukaryotes) are vital for trophic interactions and the rates of energy and nutrient flows (Sagova-Mareckova et al., 2021). In the case of lake ecosystem changes, investigating the relationships between external factors (such as rainfall, environmental factors, and nutrients) and the characteristics of microbial communities and niches can inform lake protection efforts and water environmental studies (Li et al., 2024; Liu et al., 2023b). In this study, alterations of the lake environmental heterogeneity were found to have different effects on prokaryotes and eukaryotes. Measurable responses to climatic and anthropogenic stressors in disturbed lake ecosystems can be predicted based on the changes in the microbial community (Sagova-Mareckova et al., 2021). In addition to conventional water quality parameters and algal communities, microbe-based bioindication, including both prokaryotes and eukaryotes, is suggested to be involved in water quality monitoring to reflect ecosystem functions. This study provides valuable data for microbial monitoring in Yunnan Plateau lakes. Moreover, the geographical scope needs to be expanded in further research, such as increasing the exploration of alpine lakes (Wei et al., 2023) and plain lakes (Chen et al., 2024), to investigate shifts in different types of lake ecosystems. The study of microbial succession on long time scales is an important aspect of future research through long-term monitoring or the analysis of sedimentary DNA to reveal environmental changes and consecutive lake evolution from the perspective of integrating microbial indicators (Capo et al., 2017; Li et al., 2024).

5. Conclusions

This study revealed the change trends and influencing factors of microbial community assembly and niches in plateau lakes. Remarkable decreases in Ed and increases in trophic level were observed following rainfall in the plateau lakes adjacent to cities. This indicates that the influence of runoff in the rainy season needs to be focused on to protect this type of plateau lake. We propose that nutrient runoff during the rainy season may be further managed through effective pollution interception measures (e.g., lakeside wetlands). Moreover, rainfall and environmental conditions significantly altered the microbial community assembly (through deterministic processes) and niche characteristics, which may further influence lake ecosystem functioning. To obtain a comprehensive understanding of the role of microorganisms in lake ecosystems, accurately quantifying the specific environmental conditions (such as the occurrence pattern of nutrients) that affect the microbial niche, precisely estimating the response mechanism of key species to environmental conditions, and exploring microbial community changes over a longer time scale (which is a potential limitation of this study) are also crucial. These findings extend the current understanding of the responses of microorganisms and lake ecosystems to environmental changes. Further studies are needed to investigate and reveal the ecological functions of microbes and their roles in freshwater ecosystems to provide theoretical support for the protection and ecological restoration of lakes in the future.

CRediT authorship contribution statement

Qi Liu: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Xinlu Duan:** Software, Methodology, Formal analysis. **Yang Zhang:** Methodology, Formal analysis, Data curation. **Lizeng Duan:** Methodology, Investigation, Data curation. **Xiaonan Zhang:** Software, Data curation. **Fengwen Liu:** Software, Data curation. **Donglin Li:** Formal analysis, Data curation. **Hucai Zhang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 42101151); the Yunnan Fundamental Research Projects (Nos. 202301AU070207, 202401AT070426); the NSFC-YN Project (No. U2202207); and the Special Project for Social Development of Yunnan Province (No. 202103AC100001). We thank Yun Zhang, Han Wu, and Wei Peng for their support during the fieldwork phase.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2024.119410.

References

Albert, J.S., Destouni, G., Duke-Sylvester, S.M., Magurran, A.E., Oberdorff, T., Reis, R.E., Winemiller, K.O., Ripple, W.J., 2021. Scientists' warning to humanity on the freshwater biodiversity crisis. Ambio 50 (1), 85–94.

Arar, E.J., 1997. Method 446.0: in Vitro Determination of Chlorophylls a, B, C1+ C2 and Pheopigments in Marine and Freshwater Algae by Visible Spectrophotometry. EPA, Cincinnati, Ohio.

Aslani, F., Geisen, S., Ning, D.L., Tedersoo, L., Bahram, M., 2022. Towards revealing the global diversity and community assembly of soil eukaryotes. Ecol. Lett. 25 (1), 65–76.

Baraloto, C., Hardy, O.J., Paine, C.E.T., Dexter, K.G., Cruaud, C., Dunning, L.T., Gonzalez, M.A., Molino, J.F., et al., 2012. Using functional traits and phylogenetic trees to examine the assembly of tropical tree communities. J. Ecol. 100 (3), 690–701.

Bardgett, R.D., van der Putten, W.H., 2014. Belowground biodiversity and ecosystem functioning. Nature 515 (7528), 505–511.

Bastian, M., Heymann, S., Jacomy, M., 2009. Gephi: an Open Source Software for Exploring and Manipulating Networks, pp. 361–362.

Berry, D., Widder, S., 2014. Deciphering microbial interactions and detecting keystone species with co-occurrence networks. Front. Microbiol. 5, 219.

Cai, Y.W., Wang, H.X., Zhang, T.X., Zhou, Y.W., Dong, A.Z., Huang, R., Zeng, Q.F., Yuan, H.Z., 2023. Seasonal variation regulate the endogenous phosphorus release in sediments of Shijiuhu Lake via water-level fluctuation. Environ. Res. 238.

Cao, X.F., Wang, J., Liao, J.Q., Gao, Z., Jiang, D.L., Sun, J.H., Zhao, L., Huang, Y., et al., 2017. Bacterioplankton community responses to key environmental variables in plateau freshwater lake ecosystems: a structural equation modeling and change point analysis. Sci. Total Environ. 580, 457–467.

Capo, E., Debroas, D., Arnaud, F., Perga, M.E., Chardon, C., Domaizon, I., 2017. Tracking a century of changes in microbial eukaryotic diversity in lakes driven by nutrient enrichment and climate warming. Environ. Microbiol. 19 (7), 2873–2892.

Carrara, F., Altermatt, F., Rodriguez-Iturbe, I., Rinaldo, A., 2012. Dendritic connectivity controls biodiversity patterns in experimental metacommunities. P Natl Acad Sci USA 109 (15), 5761–5766.

Cavicchioli, R., Ripple, W.J., Timmis, K.N., Azam, F., Bakken, L.R., Baylis, M., Behrenfeld, M.J., Boetius, A., et al., 2019. Scientists' warning to humanity: microorganisms and climate change. Nat. Rev. Microbiol. 17 (9), 569–586.

Chen, J., Liu, J., Han, S., Su, H., Xia, W., Wang, H., Liu, Y., Zhang, L., et al., 2023. Nontraditional biomanipulation: a powerful ecotechnology to combat cyanobacterial blooms in eutrophic freshwaters. The Innovation Life 1 (3), 100038-100031-100038-100016.

Chen, L.D., Fu, B.J., Zhang, S.R., Qiu, J., Guo, X.D., Yang, F.L., 2002. A comparative study on nitrogen-concentration dynamics in surface water in a heterogeneous landscape. Environ Geol 42 (4), 424–432.

Chen, Y.C., Yue, Y.H., Wang, J., Li, H.R., Wang, Z.K., Zheng, Z., 2024. Microbial community dynamics and assembly mechanisms across different stages of cyanobacterial bloom in a large freshwater lake. Sci. Total Environ. 907.

Chen, Z.W., Fei, Y.H., Liu, W.S., Ding, K.B., Lu, J.A., Cai, X., Cui, T.T., Tang, Y.T., et al., 2022. Untangling microbial diversity and assembly patterns in rare earth element mine drainage in South China. Water Res. 225.

Daly, R.A., Roux, S., Borton, M.A., Morgan, D.M., Johnston, M.D., Booker, A.E., Hoyt, D. W., Meulia, T., et al., 2019. Viruses control dominant bacteria colonizing the terrestrial deep biosphere after hydraulic fracturing. Nat Microbiol 4 (2), 352–361.

Delgado-Baquerizo, M., Maestre, F.T., Reich, P.B., Jeffries, T.C., Gaitan, J.J., Encinar, D., Berdugo, M., Campbell, C.D., et al., 2016. Microbial diversity drives multifunctionality in terrestrial ecosystems. Nat. Commun. 7.

Deng, Y., Jiang, Y.H., Yang, Y., He, Z., Luo, F., Zhou, J., 2012. Molecular ecological network analyses. BMC Bioinf. 13, 113.

Dini-Andreote, F., Stegen, J.C., van Elsas, J.D., Salles, J.F., 2015. Disentangling mechanisms that mediate the balance between stochastic and deterministic processes in microbial succession. P Natl Acad Sci USA 112 (11), E1326–E1332.

Dudgeon, D., 2019. Multiple threats imperil freshwater biodiversity in the Anthropocene. Curr. Biol. 29 (19), R960–R967.

Fan, Q.P., Liu, K.F., Wang, Z.L., Liu, D., Li, T., Hou, H.Y., Zhang, Z.J., Chen, D.H., et al., 2024. Soil microbial subcommunity assembly mechanisms are highly variable and intimately linked to their ecological and functional traits. Mol. Ecol. 33 (7).

Fan, X., Yang, K., Yang, R.B., Zhao, L., 2023. Changes in meteorological elements and its impacts on yunnan Plateau Lakes. Appl Sci-Basel 13 (5).

Gaston, K.J., Blackburn, T.M., Lawton, J.H., 1997. Interspecific abundance range size relationships: an appraisal of mechanisms. J. Anim. Ecol. 66 (4), 579–601.

Geng, M.D., Zhang, W.Z., Hu, T., Wang, R., Cheng, X.Y., Wang, J.J., 2022. Eutrophication causes microbial community homogenization via modulating generalist species. Water Res. 210.

Gilbert, B., 2012. Joint consequences of dispersal and niche overlap on local diversity and resource use. J. Ecol. 100 (1), 287–296.

Granot, I., Belmaker, J., 2020. Niche breadth and species richness: correlation strength, scale and mechanisms. Global Ecol. Biogeogr. 29 (1), 159–170.

Gu, Y., Li, Z.D., Lei, P., Wang, R., Xu, H., Friman, V.P., 2023. Phylogenetic distance-decay patterns are not explained by local community assembly processes in freshwater lake microbial communities. Environ. Microbiol. 25 (10), 1940–1954.

Guimera, R., Nunes Amaral, L.A., 2005. Functional cartography of complex metabolic networks. Nature 433 (7028), 895–900.

Haukka, K., Kolmonen, E., Hyder, R., Hietala, J., Vakkilainen, K., Kairesalo, T., Haari, H., Sivonen, K., 2006. Effect of nutrient loading on bacterioplankton community composition in lake mesocosms. Microb. Ecol. 51 (2), 137–146.

Hernandez, D.J., David, A.S., Menges, E.S., Searcy, C.A., Afkhami, M.E., 2021. Environmental stress destabilizes microbial networks. Isme J 15 (6), 1722–1734.

Herren, C.M., McMahon, K.D., 2017. Cohesion: a method for quantifying the connectivity of microbial communities. Isme J 11 (11), 2426–2438.

Herren, C.M., McMahon, K.D., 2018. Keystone taxa predict compositional change in microbial communities. Environ. Microbiol. 20 (6), 2207–2217.

Huber, P., Metz, S., Unrein, F., Mayora, G., Sarmento, H., Devercelli, M., 2020. Environmental heterogeneity determines the ecological processes that govern bacterial metacommunity assembly in a floodplain river system. Isme J 14 (12), 2951–2966.
Huo, S.L., Zhang, H.X., Monchamp, M.E., Wang, R., Weng, N.Y., Zhang, J.T., Zhang, H., Wu, F.C., 2022. Century-long homogenization of algal communities is accelerated by nutrient enrichment and climate warming in lakes and reservoirs of the north temperate zone. Environ. Sci. Technol. 56 (6), 3780–3790.
Ji, P.S., Wilson, M., 2002. Assessment of the importance of similarity in carbon source utilization profiles between the biological control agent and the pathogen in biological control of bacterial speck of tomato. Appl Environ Microb 68 (9), 4383–4389.
Jiao, S., Yang, Y.F., Xu, Y.Q., Zhang, J., Lu, Y.H., 2020. Balance between community assembly processes mediates species coexistence in agricultural soil microbiomes across eastern China. Isme J 14 (1), 202–216.
Kakouei, K., Kraemer, B.M., Anneville, O., Carvalho, L., Feuchtmayr, H., Graham, J.L., Higgins, S., Pomati, F., et al., 2021. Phytoplankton and cyanobacteria abundances in mid-21st century lakes depend strongly on future land use and climate projections. Global Change Biol. 27 (24), 6409–6422.
Karlsson, J., Jonsson, A., Jansson, M., 2001. Bacterioplankton production in lakes along an altitude gradient in the subarctic north of Sweden. Microb. Ecol. 42 (3), 372–382.
Kartzinel, T.R., Chen, P.A., Coverdale, T.C., Erickson, D.L., Kress, W.J., Kuzmina, M.L., Rubenstein, D.I., Wang, W., et al., 2015. DNA metabarcoding illuminates dietary niche partitioning by African large herbivores. P Natl Acad Sci USA 112 (26), 8019–8024.
Laland, K., Matthews, B., Feldman, M.W., 2016. An introduction to niche construction theory. Evol. Ecol. 30 (2), 191–202.
Legendre, P., 2014. Interpreting the replacement and richness difference components of beta diversity. Global Ecol. Biogeogr. 23 (11), 1324–1334.
Li, C.C., Wang, L.F., Ji, S.P., Chang, M.J., Wang, L.F., Gan, Y.D., Liu, J., 2021. The ecology of the plastisphere: microbial composition, function, assembly, and network in the freshwater and seawater ecosystems. Water Res. 202.
Li, Y., Zhang, H.X., Huo, S.L., Zhang, J.T., Ma, C.Z., Weng, N.Y., Zhang, P.L., Zhanyao, S., 2024. Microbial eukaryote community succession over hundreds of years in Chinese lakes. Ecol Indic 158.
Liang, Y.T., Jiang, Y.J., Wang, F., Wen, C.Q., Deng, Y., Xue, K., Qin, Y.J., Yang, Y.F., et al., 2015. Long-term soil transplant simulating climate change with latitude significantly alters microbial temporal turnover. Isme J 9 (12), 2561–2572.
Liu, Q., Chang, F.Q., Xie, P., Zhang, Y., Duan, L.Z., Li, H.Y., Zhang, X.N., Zhang, Y., et al., 2023a. Microbiota assembly patterns and diversity of nine plateau lakes in Yunnan, southwestern China. Chemosphere 314.
Liu, Q., Zhang, H., Chang, F., Xie, P., Zhang, Y., Wu, H., Zhang, X., Peng, W., et al., 2022a. eDNA revealed in situ microbial community changes in response to Trapa japonica in Lake Qionghai and Lake Erhai, southwestern China. Chemosphere 288 (3), 132605.
Liu, Q., Zhang, H.C., Chang, F.Q., Qiu, J., Duan, L.Z., Hu, G.Z., Zhang, Y., Zhang, X.N., et al., 2022b. The effect of graphene photocatalysis on microbial communities in Lake Xingyun, southwestern China. Environ Sci Pollut R 29 (32), 48851–48868.
Liu, Q., Zhang, H.C., Zhang, Y., Li, D.L., Gao, Y.H., Li, H.Y., Duan, L.Z., Zhang, X.A., et al., 2023b. Heterogeneous bacterial communities affected by phytoplankton community turnover and microcystins in plateau lakes of Southwestern China. Sci. Total Environ. 903.
Liu, W.Z., Li, S.Y., Bu, H.M., Zhang, Q.F., Liu, G.H., 2012. Eutrophication in the Yunnan Plateau lakes: the influence of lake morphology, watershed land use, and socioeconomic factors. Environ Sci Pollut R 19 (3), 858–870.
Liu, Y., Zhang, J.X., Zhao, L., Li, Y.Z., Yang, Y.Y., Xie, S.G., 2015. Aerobic and nitrite-dependent methane-oxidizing microorganisms in sediments of freshwater lakes on the Yunnan Plateau. Appl Microbiol Biot 99 (5), 2371–2381.
Ma, R.H., Yang, G.S., Duan, H.T., Jiang, J.H., Wang, S.M., Feng, X.Z., Li, A.N., Kong, F.X., et al., 2011. China's lakes at present: number, area and spatial distribution. Sci. China Earth Sci. 54 (2), 283–289.
Manzoni, S., Schimel, J.P., Porporato, A., 2012. Responses of soil microbial communities to water stress: results from a meta-analysis. Ecology 93 (4), 930–938.
Massana, R., Logares, R., 2013. Eukaryotic versus prokaryotic marine picoplankton ecology. Environ. Microbiol. 15 (5), 1254–1261.
Montoya, J.M., Pimm, S.L., Sole, R.V., 2006. Ecological networks and their fragility. Nature 442 (7100), 259–264.
Ngoye, E., Machiwa, J.F., 2004. The influence of land-use patterns in the Ruvu river watershed on water quality in the river system. Phys. Chem. Earth 29 (15–18), 1161–1166.
Noges, T., 2009. Relationships between morphometry, geographic location and water quality parameters of European lakes. Hydrobiologia 633 (1), 33–43.
Ofek-Lalzar, M., Sela, N., Goldman-Voronov, M., Green, S.J., Hadar, Y., Minz, D., 2014. Niche and host-associated functional signatures of the root surface microbiome. Nat. Commun. 5.
Östman, Ö., Drakare, S., Kritzberg, E.S., Langenheder, S., Logue, J.B., Lindström, E.S., 2012. Importance of space and the local environment for linking local and regional abundances of microbes. Aquat. Microb. Ecol. 67 (1), 35–U158.
Parker, B.R., Vinebrooke, R.D., Schindler, D.W., 2008. Recent climate extremes alter alpine lake ecosystems. P Natl Acad Sci USA 105 (35), 12927–12931.
Podani, J., Schmera, D., 2011. A new conceptual and methodological framework for exploring and explaining pattern in presence-absence data. Oikos 120 (11), 1625–1638.
Ran, J., Xiang, R., He, J., Zheng, B.H., 2023. Spatiotemporal variation and driving factors of water quality in Yunnan-Guizhou plateau lakes, China. J. Contam. Hydrol. 254.
Ranjard, L., Dequiedt, S., Prevost-Boure, N.C., Thioulouse, J., Saby, N.P.A., Lelievre, M., Maron, P.A., Morin, F.E.R., et al., 2013. Turnover of soil bacterial diversity driven by wide-scale environmental heterogeneity. Nat. Commun. 4.
Ren, C.J., Chen, J., Lu, X.J., Doughty, R., Zhao, F.Z., Zhong, Z.K., Han, X.H., Yang, G.H., et al., 2018. Responses of soil total microbial biomass and community compositions to rainfall reductions. Soil Biol. Biochem. 116, 4–10.
Ren, Z., Qu, X.D., Zhang, M., Yu, Y., Peng, W.Q., 2019. Distinct bacterial communities in wet and dry seasons during a seasonal water level fluctuation in the largest freshwater lake (Poyang Lake) in China. Front. Microbiol. 10.
Sagova-Mareckova, M., Boenigk, J., Bouchez, A., Cermakova, K., Chonova, T., Cordier, T., Eisendle, U., Elersek, T., et al., 2021. Expanding ecological assessment by integrating microorganisms into routine freshwater biomonitoring. Water Res. 191.
Schindler, D.W., 2006. Recent advances in the understanding and management of eutrophication. Limnol. Oceanogr. 51 (1), 356–363.
She, Z.X., Pan, X., Wang, J., Shao, R., Wang, G.C., Wang, S.P., Yue, Z.B., 2021. Vertical environmental gradient drives prokaryotic microbial community assembly and species coexistence in a stratified acid mine drainage lake. Water Res. 206.
Shen, C.C., Gunina, A., Luo, Y., Wang, J.J., He, J.Z., Kuzyakov, Y., Hemp, A., Classen, A. T., et al., 2020. Contrasting patterns and drivers of soil bacterial and fungal diversity across a mountain gradient. Environ. Microbiol. 22 (8), 3287–3301.
Shen, M.Y., Li, Q., Ren, M.L., Lin, Y., Wang, J.P., Chen, L., Li, T., Zhao, J.D., 2019. Trophic status is associated with community structure and metabolic potential of planktonic microbiota in Plateau Lakes. Front. Microbiol. 10.
Sommaruga, R., 2001. The role of solar UV radiation in the ecology of alpine lakes. J Photoch Photobio B 62 (1–2), 35–42.
Spanbauer, T.L., Allen, C.R., Angeler, D.G., Eason, T., Fritz, S.C., Garmestani, A.S., Nash, K.L., Stone, J.R., et al., 2016. Body size distributions signal a regime shift in a lake ecosystem. P Roy Soc B-Biol Sci 283 (1833).
State Environmental Protection Administration of China, 2002. Water and Wastewater Monitoring and Analysis Methods, fourth ed. China Environmental Science Press, Beijing, p. 783 (in Chinese).
Stegen, J.C., Lin, X.J., Konopka, A.E., Fredrickson, J.K., 2012. Stochastic and deterministic assembly processes in subsurface microbial communities. Isme J 6 (9), 1653–1664.
Su, W.H., Han, Q., Yang, J.W., Yu, Q.L., Wang, S.J., Wang, X.C., Qu, J.P., Li, H., 2022. Heavy rainfall accelerates the temporal turnover but decreases the deterministic processes of buried gravesoil bacterial communities. Sci. Total Environ. 836.
Sun, W.J., Jing, Z.B., 2023. Migration of rare and abundant species, assembly mechanisms, and ecological networks of microbiomes in drinking water treatment plants: effects of different treatment processes. J. Hazard Mater. 457.
Wan, W., Grossart, H.P., He, D., Liu, W., Wang, S., Yang, Y., 2023. Differentiation strategies for planktonic bacteria and eukaryotes in response to aggravated algal blooms in urban lakes. iMeta 2 (1), e84.
Wang, H.H., Ke, H.Y., Wu, H.P., Ma, S.Y., Altaf, M.M., Diao, X.P., 2024. Season shapes the functional diversity of microbial carbon metabolism in mangrove soils of Hainan Island, China. Catena 235.
Wang, J.L., Fu, Z.S., Qiao, H.X., Liu, F.X., 2019. Assessment of eutrophication and water quality in the estuarine area of Lake Wuli, Lake Taihu, China. Sci. Total Environ. 650, 1392–1402.
Wang, S.H., Yuan, S.W., Che, F.F., Wan, X., Wang, Y.F., Yang, D.H., Yang, H.J., Zhu, D., et al., 2023. Strong bacterial stochasticity and fast fungal turnover in Taihu Lake sediments, China. Environ. Res. 237.
Webb, C.O., Ackerly, D.D., McPeek, M.A., Donoghue, M.J., 2002. Phylogenies and community ecology. Annu Rev Ecol Syst 33, 475–505.
Wei, C., Sun, D., Yuan, W.L., Li, L., Dai, C.X., Chen, Z.Z., Zeng, X.M., Wang, S.H., et al., 2023. Metagenomics revealing molecular profiles of microbial community structure and metabolic capacity in Bamucuo lake, Tibet. Environ. Res. 217.
Wisnoski, N.I., Lennon, J.T., 2023. Scaling up and down: movement ecology for microorganisms. Trends Microbiol. 31 (3), 242–253.
Wong, W.H., Dudula, J.J., Beaudoin, T., Groff, K., Kimball, W., Swigor, J., 2018. Declining ambient water phosphorus concentrations in Massachusetts' rivers from 1999 to 2013: environmental protection works. Water Res. 139, 108–117.
Woolway, R.I., Merchant, C.J., 2019. Worldwide alteration of lake mixing regimes in response to climate change. Nat. Geosci. 12 (4), 271–276.
Wu, H., Zhang, H.C., Zhang, X.N., Peng, W., Liu, Q., Liu, F.W., Zhang, Y., Duan, L.Z., 2021. Potential catastrophic water outflow from Lake Dian, China: possible hydrological and ecological risks. Catena 207.
Xiong, X., Wang, W.L., Xing, Y.F., Chen, H.Z., Luo, X.S., Chen, W.L., Huang, Q.Y., 2022. Niche overlap is a predictor of the interspecies correlations detected by microbial network analysis in soil micro-aggregates. J Soil Sediment 22 (5), 1521–1529.
Xu, C., Wan, R.R., Yang, G.S., Lv, W., Ma, Q., Li, B., 2019. Water quality control plan with BATHTUB model for lake inflow rivers - a case study of total phosphorus in Northwest Lake Taihu, China. Water Sci. Technol. 79 (3), 544–555.
Yang, N., Hou, X., Li, Y., Zhang, H.J., Wang, J., Hu, X.D., Zhang, W.L., 2023. Inter-basin water diversion homogenizes microbial communities mainly through stochastic assembly processes. Environ. Res. 223.
Zhang, J.X., Yang, Y.Y., Zhao, L., Li, Y.Z., Xie, S.G., Liu, Y., 2015. Distribution of sediment bacterial and archaeal communities in plateau freshwater lakes. Appl Microbiol Biot 99 (7), 3291–3302.
Zhang, L., Xu, K.C., Wang, S.R., Wang, S.G., Li, Y.P., Li, Q.C., Meng, Z., 2018. Characteristics of dissolved organic nitrogen in overlying water of typical lakes of Yunnan Plateau, China. Ecol Indic 84, 727–737.
Zhang, W.P., Ding, W., Li, Y.X., Tam, C., Bougouffa, S., Wang, R.J., Pei, B.T., Chiang, H. Y., et al., 2019. Marine biofilms constitute a bank of hidden microbial diversity and functional potential. Nat. Commun. 10.

Zhang, X., Xie, P., Chen, F.Z., Li, S.X., Qin, J.H., 2007. Driving forces shaping phytoplankton assemblages in two subtropical plateau lakes with contrasting trophic status. Freshwater Biol 52 (8), 1463–1475.

Zhang, X.N., Zhang, H.C., Chang, F.Q., Xie, P., Li, H.Y., Wu, H., Ouyang, C.T., Liu, F.W., et al., 2021a. Long-range transport of aeolian deposits during the last 32 kyr inferred from rare earth elements and grain-size analysis of sediments from Lake Lugu, Southwestern China. Palaeogeogr Palaeocl 567.

Zhang, Y., Dong, S., Gao, Q., Liu, S., Zhou, H., Ganjurjav, H., Wang, X., 2016. Climate change and human activities altered the diversity and composition of soil microbial community in alpine grasslands of the Qinghai-Tibetan Plateau. Sci. Total Environ. 562, 353–363.

Zhang, Y., Zhang, H., Chang, F., Xie, P., Liu, Q., Duan, L., Wu, H., Zhang, X., et al., 2021b. In-situ responses of phytoplankton to graphene photocatalysis in the eutrophic lake Xingyun, southwestern China. Chemosphere 278, 130489.

Zhang, Y., Zhang, H.C., Liu, Q., Duan, L.Z., Zhou, Q.C., 2023. Total nitrogen and community turnover determine phosphorus use efficiency of phytoplankton along nutrient gradients in plateau lakes. J Environ Sci 124, 699–711.

Zhou, J.Z., Deng, Y., Zhang, P., Xue, K., Liang, Y.T., Van Nostrand, J.D., Yang, Y.F., He, Z. L., et al., 2014. Stochasticity, succession, and environmental perturbations in a fluidic ecosystem. P Natl Acad Sci USA 111 (9), E836–E845.

Zhou, J.Z., Ning, D.L., 2017. Stochastic community assembly: does it matter in microbial ecology? Microbiol Mol Biol R 81 (4).

Zinger, L., Taberlet, P., Schimann, H., Bonin, A., Boyer, F., De Barba, M., Gaucher, P., Gielly, L., et al., 2019. Body size determines soil community assembly in a tropical forest. Mol. Ecol. 28 (3), 528–543.